Remanent Magnetization of the Upper-Miocene 'Blue' Sandstones of California

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Abstract—A study has been made of the magnetization of some upper-Miocene sediments of California. Field evidence and laboratory experiments indicate that the main component of magnetization of these sediments was not acquired by alignment of detrital magnetic grains at the time of deposition, nor is it due to a later isothermal magnetization of these grains. A strong and stable magnetization appears to have developed through post-depositional and, in some cases, post-deformational growth of crystalline magnetic minerals *in situ*.

Introduction—Since this report is concerned with remanent magnetization of sedimentary rocks, it is desirable to discuss briefly how such rocks receive a remanent magnetization. *Nagata* [1953] has named the remanent magnetization that a rock possesses in the field 'natural remanent magnetization' (NRM). NRM is, in general, composed of two components, 'thermo-remanent magnetization' (TRM) and 'isothermal-remanent magnetization' (IRM). As will be proposed in this report, there may also be a third component, termed here 'crystallization magnetization.'

Magnetic minerals—essentially magnetite, hematite, maghemite, titanomagnetites, and some iron sulfides (likewise the rocks containing them)—acquire TRM by being heated above their Curie points and then cooling in a magnetic field, which may be relatively weak (of the order of the Earth's field, about 0.5 oe). IRM is, on the other hand, imparted to a rock at room temperature by the application of an external field (usually large); that is, the rock may show the well-known property of hysteresis. Both IRM and TRM are normally aligned in the direction of the field acting at the time they are imparted to the rock. It is apparent, then, that igneous rocks will acquire a TRM when they cool in the Earth's magnetic field. They may also, at a later date, acquire an IRM from lightning bolts, or possibly from the influence of the Earth's field acting over long periods of time.

Sedimentary rocks are, of course, formed from material primevally eroded from igneous rocks. Consequently, the magnetic mineral grains they contain may possess TRM or IRM which was originally imparted to the grain in the parent igneous rock, or both. During the process of sedimentation these grains may become statistically aligned in the Earth's magnetic field [*Clegg* and others, 1954; *Johnson* and others, 1948; *Nagata* and others, 1943] and thus give the sediment a remanent magnetization. Further, the sediment may acquire a component of IRM or crystallization magnetization after consolidation, or both.

Of further interest here are the properties of IRM and TRM. The principal difference between these two types of magnetization lies in their respective stabilities. TRM is but slightly changed by the application of many disturbing effects, and is destroyed only by heating the material to its Curie point. IRM is, on the other hand, considerably changed by the application of ac magnetic fields [*Matsusaki* and others, 1954] or by heating to relatively low temperatures.

Geologic setting—This report describes experiments performed on sandstones of uppermost-Miocene age occurring at three localities in the San Francisco Bay region of California. These rocks are very similar in lithology to the Etchegoin formation, occurring in the Great Valley of California. *Lynton* [1931] found some rather strong magnetic anomalies over the Kettleman Hills oil field which he attributed to the Etchegoin formation. Because of the apparently strong magnetic character of these sediments, they were selected for remanent magnetic studies.

Twelve oriented samples were taken from Corral Hollow, east of Livermore, California, in the summer of 1953. Subsequently, seven samples were collected in Bailey Canyon, just south of Pittsburg, California, and ten samples were taken from two localities east of Santa Cruz, California. The locations of these three areas are shown in Figure 1.

The beds in Corral Hollow, the Neroly formation [*Huey*, 1948], are folded into a rather steep syncline and anticline. A map showing the location of the samples taken here is given in Figure 2, with a geologic cross-section of these beds. Grain sizes of the sandstones vary from coarse (1–2 mm in diameter) to very fine siltstones, and some clays are present. Some of the rocks are well cemented with carbonates and others are practically devoid of cement. The most characteristic

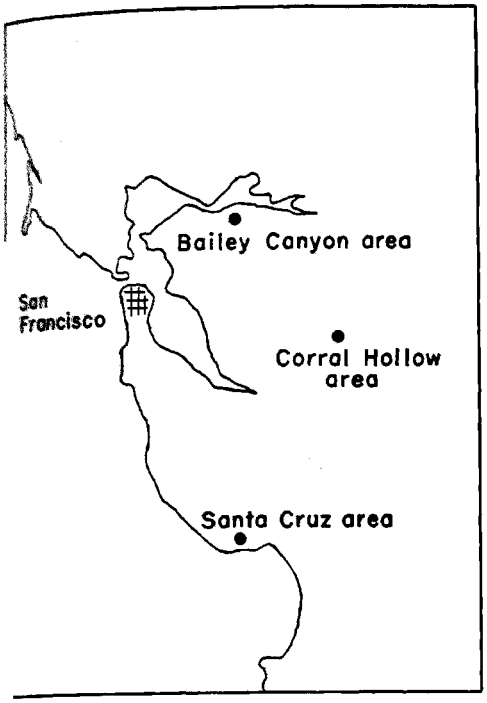


FIG. 1 – Location map

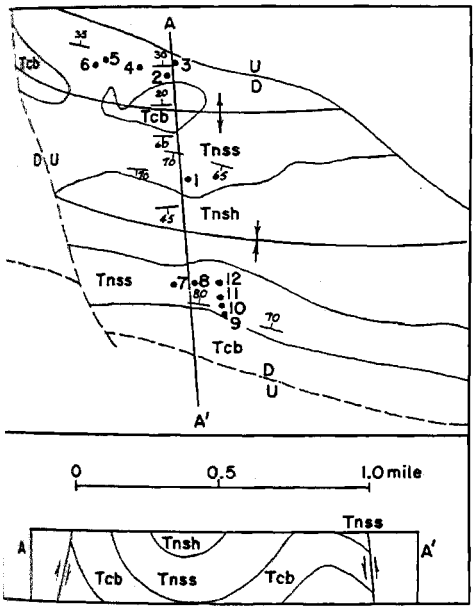


FIG. 2 – Sample locations at Corral Hollow: Tnsh—Upper Neroly (shales); Tnss—Lower Neroly (blue sandstones); Tcb—Cierbo (sandstone)

feature of these sandstones, as well as those from the other two localities, is a distinctive blue color seen in the hand specimen. The color is due to a very thin coating (0.007 mm average thickness)

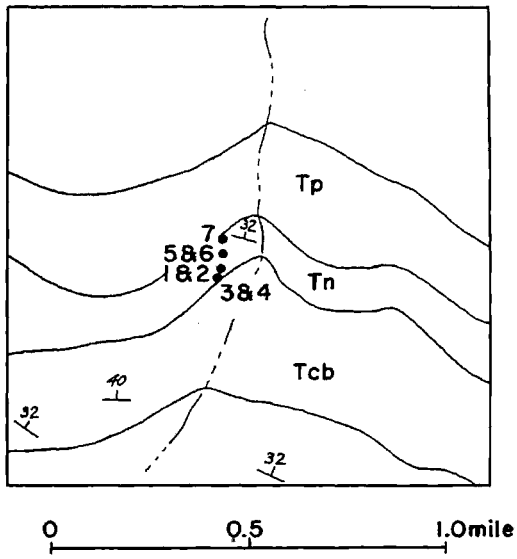


FIG. 3 – Sample locations at Bailey Canyon: Tp—Pinole Tuff; Tn—Neroly (undifferentiated); Tcb—Cierbo

of translucent material which covers nearly all grains, but does not fill the interstices. The coating is colorless when seen in thin section, but when covering the dark grains in the rock, it gives rise to the distinctive blue color. J. Lerbeckmo (personal communication, 1955), on the basis of x-ray diffraction patterns and other tests, finds the coating to be an iron-rich clay material with a montmorillonite-type structure, although it was previously thought to be opal [*Patten*, 1947]. The age of the Neroly is given as uppermost-Miocene [*Huey*, 1948].

In Bailey Canyon, the Neroly also occurs in a section with other Tertiary sediments, all of which are dipping uniformly to the north at about 35°. The geologic map shown in Figure 3, with the sample locations, is taken from *Patten* [1947]. Lithology is practically identical with that at Corral Hollow.

In the Santa Cruz area the blue sandstones are flat-lying and apparently have been undisturbed since deposition. Six samples were taken near the eastern edge of Section 21, R.1 W., T.10 S., New Almaden Quadrangle, and four were taken in Section 14, R.1 W., T.10 S., New Almaden Quadrangle. The samples were taken from different stratigraphic horizons and were usually separated by a few hundred feet horizontally. The lithology is very similar to that of the Neroly sandstones, except for a general lack of cementing. On the basis of detrital mineral grain analyses, *McAndrews* [1948] has correlated these rocks with the

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TABLE 1 – *Magnetization of specimens*

No.	Azimuth of magnetic axis	Inclination of magnetic axis (down)	Intensity of magnetization $\times 10^6$	Geologic attitude	
				Strike	Dip
	°	°	cgsu/cc	°	°
B-1-a	N16E	46	15.5	S80E	62S
B-1-b	N15E	46	16.0	S80E	62S
B-2-a	N26E	53	24.5	S50E	20N
B-3-a	N5W	67	36.3	S70E	27N
B-3-b	N5W	66	35.5	S70E	27N
B-3-c	N1W	65	36.0	S70E	27N
B-4-a	DueN	59	31.1	S40E	30N
B-4-b	N4E	56	31.3	S40E	30N
B-6-1	N21E	60	37.4	N45W	20N
B-6-2	N16E	60	38.6	N45W	20N
B-7-a	N4W	65	22.5	S70E	90
B-7-b	N6E	65	21.6	S70E	90
B-8-1	N13W	68	20.1	N60W	80N
B-9-a	N17E	81	22.2	S65E	80N
B-9-b	N4W	76	23.2	S65E	80N
B-10-a	N32E	58	33.0	S65E	80N
B-10-b	N22E	64	28.2	S65E	80N
B-10-c	N11E	61	21.1	S65E	80N
B-10-d	N27E	63	21.1	S65E	80N
B-10-e	N5W	66	27.8	S65E	80N
B-11-a	N10E	62	32.4	S65E	80N
B-11-b	N17E	66	31.3	S65E	80N
B-11-c	N15E	65	28.2	S65E	80N
B-12-1	N11E	62	35.3	N75W	78N
BC-1-1	N19E	56	13.0	N72W	35N
BC-1-2	N20E	52	12.6	N72W	35N
BC-2-1	N2E	55	5.46	N72W	35N
BC-2-2	N8E	59	5.60	N72W	35N
BC-2-3	N4W	56	6.92	N72W	35N
BC-3-1	DueN	60	17.8	N85W	36N
BC-3-2	N14E	66	19.0	N85W	36N
BC-4-1	N12E	55	18.5	N85W	36N
BC-4-2	N3W	56	15.5	N85W	36N
BC-5-1	N15E	55	30.8	E-W	21N
BC-5-2	N13E	55	29.1	E-W	21N
BC-6-a	N12W	58	5.98	E-W	21N
BC-7-a	N27E	61	6.64	E-W	29N
SC-1-1	N20E	50	15.5	N12W	8W
SC-1-2	N20E	57	10.7	N12W	8W
SC-1-3	N24E	58	9.57	N12W	8W
SC-3-1	N31W	70	8.20	N75E	18S
SC-5-1	N3W	57	8.22	Horizontal	
SC-5-2	N8E	73	7.70	Horizontal	
SC-5-3	N8W	63	8.62	Horizontal	
SC-6-1	S46W	44	8.46	Horizontal	
SC-6-2	S25W	45	8.98	Horizontal	
SC-7-1	N17E	52	7.60	Horizontal	
SC-7-2	N8E	60	9.82	Horizontal	
SC-7-3	N15E	62	9.60	Horizontal	
SC-8-1	N10E	57	3.03	Horizontal	
SC-8-2	N32E	57	2.68	Horizontal	
SC-8-3	N14E	53	2.74	Horizontal	
SC-9-1	N24W	48	2.37	Horizontal	
SC-9-2	N9E	31	3.58	Horizontal	
SC-9-3	N23W	43	2.95	Horizontal	
SC-9-4	N19W	36	2.67	Horizontal	
SC-10-1	N8W	55	5.76	Horizontal	
SC-10-2	N1W	54	5.95	Horizontal	

Neroly formation, and placed them as uppermost Miocene.

The heavy mineral assemblages of the rocks from all three areas are characterized by abundant hypersthene and augite, with hornblende in somewhat lesser amounts. *Huey* [1948], *Patten* [1947], and *McAndrews* [1948] all suggest andesites and basalts as the source rocks for these blue sandstones, presumably the volcanic rocks of the Sierra Nevada. It is therefore probable that the blue sandstones of these three areas were deposited at the same time and were derived from a common source.

Experimental—The 29 oriented samples collected at the various localities were formed into 58 one-inch cubes for measurement of their remanent magnetization. From one to five cubes were cut from each sample. Measurements were made on a 'rock-generator' type magnetometer [*Johnson*, 1938], which was designed and constructed at the University of California. Details of the equipment and its operation, methods used in sampling and preparing specimens, and the reduction of the experimental data to direction and intensity of magnetization of the specimens are given elsewhere [*Doell*, 1955].

For specimens having moments of 5×10^{-3} units (cgs em) or larger, the probable error in a single determination of direction is estimated at 5°. This is due to errors in orientation, about 2–3°, errors in cubing, 2–3°, and errors in measurement of 2°.

Direction of magnetization—The direction of magnetization and intensity of magnetization of these specimens are given in Table 1, along with the strike and dip of the beds from which the samples were taken. The directions of magnetization, as given in this table, are not corrected for geologic dip. Samples collected from Corral Hollow are prefixed by the letter B, those from Bailey Canyon by BC, and those from the Santa Cruz area by SC. The last digit or lower-case letter appearing in the number is the number of the specimen cut from any one sample.

The directions of magnetization of these samples are given in graphical form, by means of the Schmidt equal-area projection, in Figures 4, 5, and 6 for the Corral Hollow, Bailey Canyon, and Santa Cruz areas, respectively. All of the directions of magnetization measured were north-seeking downward; thus, all points on the Schmidt projections are on the lower hemisphere.

Since these directions are not corrected for geologic dip, it is apparent that the magnetization was acquired after the beds were folded, that is

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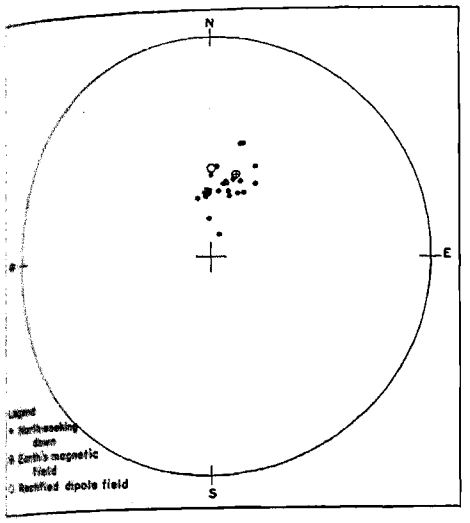


FIG. 4 - Directions of remanent magnetization at Corral Hollow

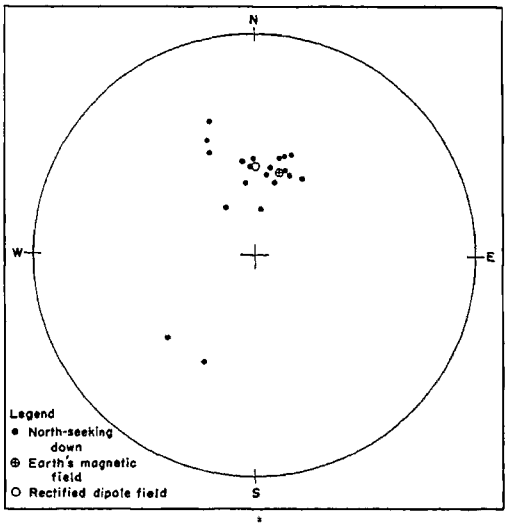


FIG. 6 - Directions of remanent magnetization at Santa Cruz

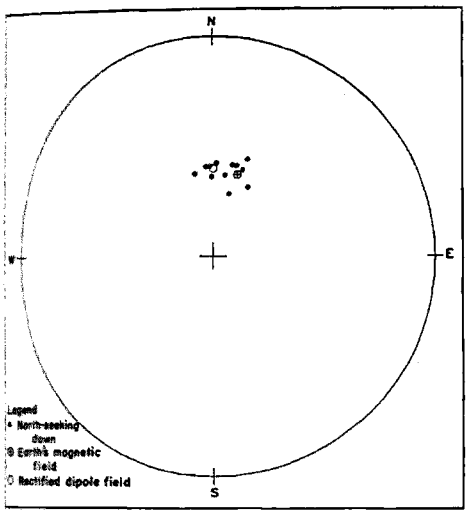


FIG. 5 - Directions of remanent magnetization at Bailey Canyon

for the Corral Hollow area (and also for the Bailey Canyon area, as will be pointed out later).

In measurements of the direction of magnetization of a number of specimens from some particular series of rocks, it is always of interest to determine the average direction of magnetization and the significance of this average direction. The statistical treatment of dispersion on a sphere [*Fisher*, 1953] is used for this test of significance. By this method, the average azimuth and inclination of the magnetic axes of the specimens are determined along with an 'angle of confidence,' θ. This is the semi-angle of a cone whose apex is at the geographical point in question and

TABLE 2 - *Average directions of magnetization and 'angles of confidence'*

Area	Azimuth E. of N.	Inclination	θ
	°	°	°
Corral Hollow.....	10½	63	3¾
Bailey Canyon.....	8½	58½	3½
Santa Cruz.....	3	56	6¼
All areas.....	7	58	2¾

whose axis is the direction given by the average azimuth and inclination. For a 95-pct probability, the true direction of magnetization lies within this cone for 95 pct of such determinations. Any value of probability may be used, and 95 pct is used throughout this report. This treatment was applied to the data in Table 1 with the following results, shown in Table 2.

The two data from the Santa Cruz area in the SW quadrant (the two specimens cut from sample SC-6) were not used in this calculation. Apart from the possibility of misorientation (they are roughly 90° from the average direction), no reason can be given for this large discrepancy.

These results are shown in graphical form in Figure 7. Also indicated on the projection are the directions of the present geomagnetic field in the area, and of the rectified dipole field. The latter is that which would be due to a magnetic dipole situated at the center of the Earth and oriented along the axis of rotation from north to south. This is considered to be the average direction of the Earth's magnetic field when taken over a few thousand years [*Hospers*, 1955]. The average

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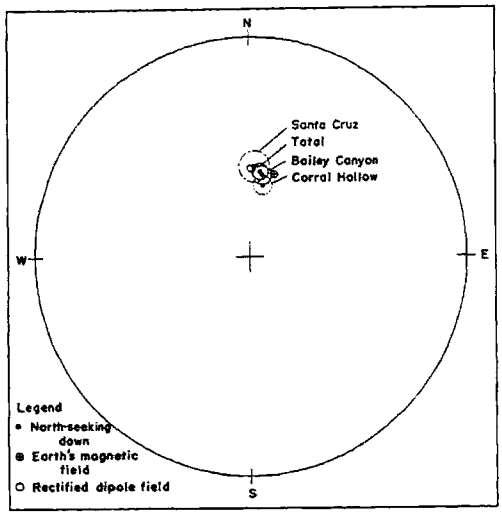


FIG. 7 – Average directions with 'circles of confidence'

directions calculated from the three separate areas are not significantly different from each other, but are significantly different from the Earth's present magnetic field, which in this area has a declination of 18° E and an inclination of 58°. The rectified dipole field is inclined at 57°.

The magnetization of the Corral Hollow rocks is obviously not that which they may have acquired at the time they were deposited. If the magnetization had been acquired during deposition, and had retained its original direction, then the subsequent folding should have considerably scattered the directions of the individual samples. In such a case, by correcting the directions of magnetization for geologic dip, that is, 'unfolding the beds,' this scatter would be decreased and the directions would all approach the average direction [*Graham*, 1949]. In the Corral Hollow rocks, the individual directions cluster about the average direction *in situ*, and correcting for geologic dip causes considerable scatter. Therefore, these rocks received a 'uniform' magnetization after they were folded. Further, the magnetization of the Corral Hollow rocks may not have been acquired over a period of time of several thousand years under the action of the Earth's field. This assumes that the field acting at the time these rocks received their magnetization had, on the average, the same direction as the rectified dipole field. *Hospers* [1955] concluded that this has been the case since the Oligocene.

The same conclusions can likewise be drawn from the measurements of the Bailey Canyon rocks, that is, that they received their magnetization after they were tilted, and that this magnetization may have been acquired during a relatively short period of time. The scatter in the data used by Hospers, however, does not completely exclude the possibility of slight long-term departures of the average field from the direction of the rectified dipole field since that time. The Bailey Canyon and Corral Hollow rocks probably received their magnetization only very recently, that is, after folding (lower Pleistocene). Thus, the magnetization of these rocks (if acquired over a period of several thousand years) may indicate a slight departure of the direction of the average field from that of the rectified dipole field during the lower Pleistocene. It may be, however, that these magnetizations are not stable over periods of time of the order of a few thousand years, and are, therefore, altered by the present field of the Earth.

The results for the Santa Cruz area are what might be expected for a stable magnetization that had been acquired at the time of deposition. Since the samples surely cover a period of deposition of at least several thousand years, the average direction of magnetization of such a group of samples should be that of the average field which existed during the period of deposition. There is, of course, no way of determining from the measurements of the directions of magnetization whether the magnetization of the Santa Cruz rocks is or is not completely stable. However, stability is indicated by the laboratory tests and the significant difference of the average direction from the direction of the geomagnetic field. Moreover, it is unlikely that these rocks owe their magnetization to an alignment of the detrital magnetic grains (see below).

It is worth noting that the period of deposition of these sediments (uppermost-Miocene) is a period for which reversals of the main geomagnetic field have been postulated [*Roche*, 1953]. Late Pliocene (possibly lower Pleistocene) sediments are folded with the Neroly, whereas middle Pleistocene sediments are not (C. M. Gilbert, personal communication, 1955). Thus the main period of folding occurred during the lower Pleistocene. This is also reported as a period of reversals [*Hospers*, 1955]. Although the Neroly sediments apparently no longer reflect the direction of the geomagnetic field at the time of their deposition (if they ever did), it is thought that they were magnetized under the action of the Earth's field during the later phases of deformation or just after deformation (that is, lower Pleistocene). All of the rocks reported here are 'normally' magnetized.

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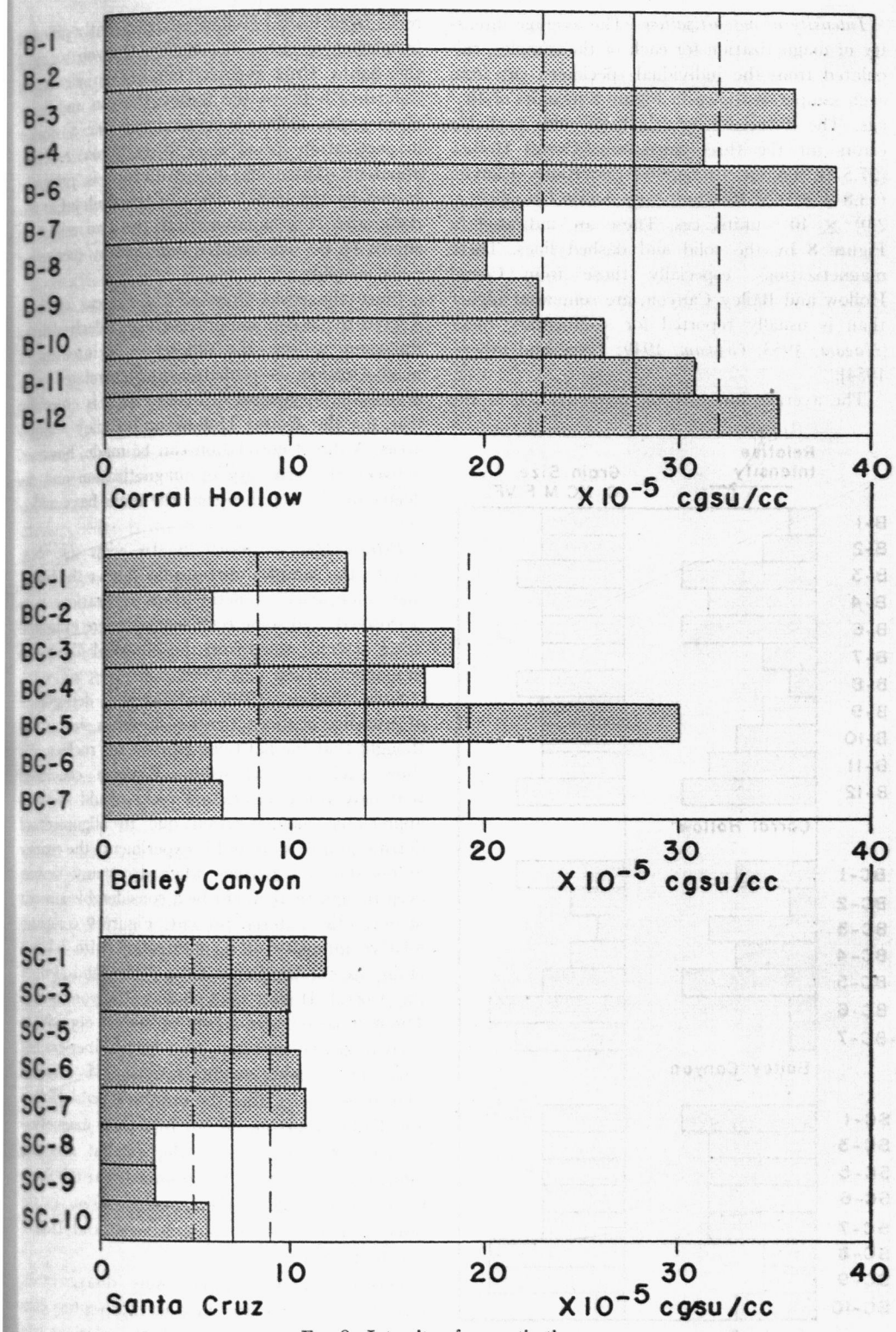


FIG. 8 – Intensity of magnetization

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Intensity of magnetization—The average intensity of magnetization for each of the samples, calculated from the individual specimens cut from each sample, is shown in Figure 8 in units of 10^{-5} cgs. The average magnetizations with probable errors for the three areas are: Corral Hollow $(27.5 \pm 4.6) \times 10^{-5}$ units, cgs; Bailey Canyon $(13.8 \pm 5.5) \times 10^{-5}$ units, cgs; Santa Cruz $(7.2 \pm 2.0) \times 10^{-5}$ units, cgs. These are indicated in Figure 8 by the solid and dashed lines. These magnetizations, especially those from Corral Hollow and Bailey Canyon, are somewhat higher than is usually reported for sedimentary rocks [*Nagata*, 1953; *Graham*, 1949; *Clegg* and others, 1954].

The average intensity of magnetization of the rocks from Corral Hollow is significantly different from that of either the Bailey Canyon rocks or the Santa Cruz rocks. There is no significant difference between the Bailey Canyon and Santa Cruz areas, although a tendency for a weaker magnetization from the Santa Cruz rocks is apparent. This lack of significance is primarily caused by the large scatter of the Bailey Canyon data, and it is possible that the two might be shown to be significantly different by measuring more samples.

Since the rocks from all three areas are considered to be contemporaneous and derived from the same source, the difference in intensity of magnetization is probably not correlated with differences in general lithology. This is especially true for the Corral Hollow and Bailey Canyon areas. A direct correlation can be made, however, between the intensity of magnetization and the degree of diastrophism that the rocks have undergone.

Petrographic examination—In view of these results, the samples were studied under the binocular microscope, mineral grain separations were performed, and some rock sections were examined in an attempt to explain the observed differences in magnetization.

Since *Graham* [1949] had found a dependence of intensity of magnetization on grain size, it was thought that the flat-lying Santa Cruz rocks might show such a relationship. Graham concluded that only the finer-grained rocks would show an appreciable magnetization due to alignment of detrital grains, since in his experiments the coarser grains did not become aligned to any degree even though there might be a considerable amount of magnetic material present. Figure 9 compares relative intensity of magnetization with relative grain size for the Santa Cruz rocks (and also for the Corral Hollow and Bailey Canyon rocks). The correlation here seems to be just opposite to what might be expected, that is, the finer-grained rocks show weaker magnetizations. This indicates that these rocks, like those from Corral Hollow and Bailey Canyon, do not owe their magnetization to an alignment of the detrital magnetic mineral grains. Thus, it is assumed that the Santa Cruz rocks received their magnetization in the same manner as did those from Corral Hollow and Bailey Canyon.

Portions of the samples were disaggregated, formed into composite samples representing each area, washed in acidic and basic solutions (the basic solution was required in order to remove the

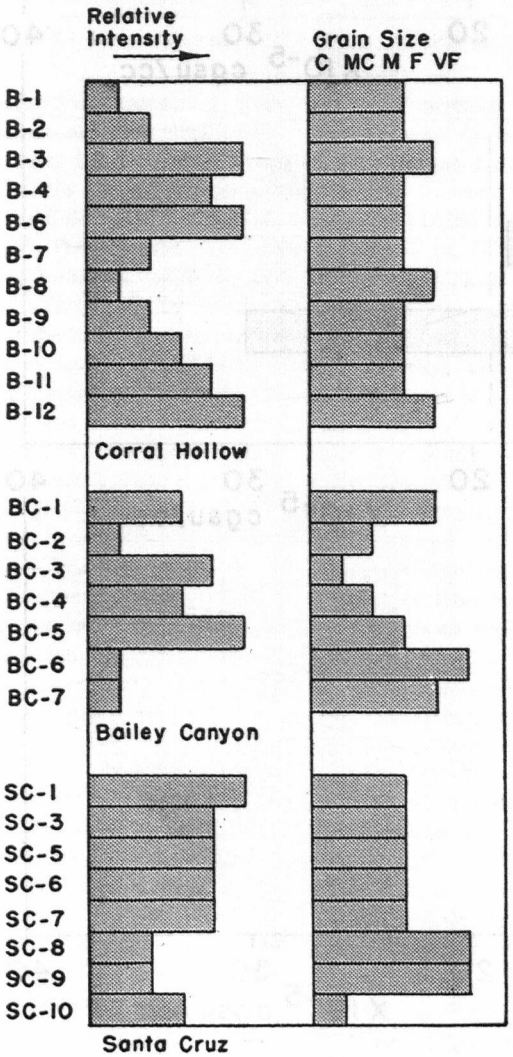


FIG. 9 – Grain size and relative intensity

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thin clay coating) and separated into light and heavy fractions by use of bromoform [*Milner*, 1940]. The 'magnetic' grains were then removed from the heavy fraction by the use of a small hand magnet. Examination of these grains under the binocular microscope reveals that most of them are about 0.1 to 0.5 mm in diameter, and are composed of a glassy material containing inclusions of an opaque material, presumably magnetite or a related mineral. The opaque material constitutes up to 50 pct of the grain in a few cases, but normally forms a much smaller portion of the grain. Smaller opaque grains, averaging about 0.05 mm in diameter, are also present in lesser amounts (perhaps two to three per cent of the total 'magnetic' grains). Most of this material shows effects of transport and is undoubtedly derived from the volcanic source rocks. It is extremely unlikely that such grains would receive a statistical alignment in the Earth's field. Further, there is no observable difference in the 'magnetic' grains from the different areas.

X-ray powder diffraction patterns were made from this material identifying it as magnetite having a cell dimension of about 8.40 A. There is no significant difference in the lattice dimensions of grains from the different areas.

Spectrochemical determinations were also made on this material, to test for differences between the areas. The results of these determinations, which were carried out by G. M. Gordon at the Division of Mineral Technology, University of California, Berkeley, California, as tests number 629-55 and 616-55 are shown in Table 3. Small amounts of *V*, *Cr*, *CaO*, *MgO*, etc., are also present in all samples. From these tests it is concluded that the detrital magnetic mineral grains of the three areas are made up of the same material.

Although the character of the 'magnetic' grains makes it difficult to determine relative percentages of magnetite from the rocks at the various localities, there appear to be only slight differences. Thus, the differences in intensity of magnetization of the three areas are apparently not correlated with differences in the detrital magnetic mineral grains.

Chemical activity—There is, in all these rocks, an indication of post-depositional chemical activity involving iron. The deposition of the iron-rich clay coating on the grains indicates the previous presence of iron-rich solutions. Serrated edges on many hypersthene grains indicate solution of these minerals and therefore a liberation of some iron. It was noted, however, that the etching of the hypersthene grains occurred after deposition of the iron-rich clay coating. In some thin sections of the Neroly formation (obtained from J. Lerbeckmo) an opaque material is observed that fills or partially fills some of the inter-

TABLE 3 – *Results of spectrochemical analyses*

Sample	SiO_2	TiO_2	Al_2O_3	Mn
	pct	pct	pct	pct
B	10	1.0	1.0	0.1
BC	5	1.2	1.0	0.1
SC	10	1.2	1.0	0.2

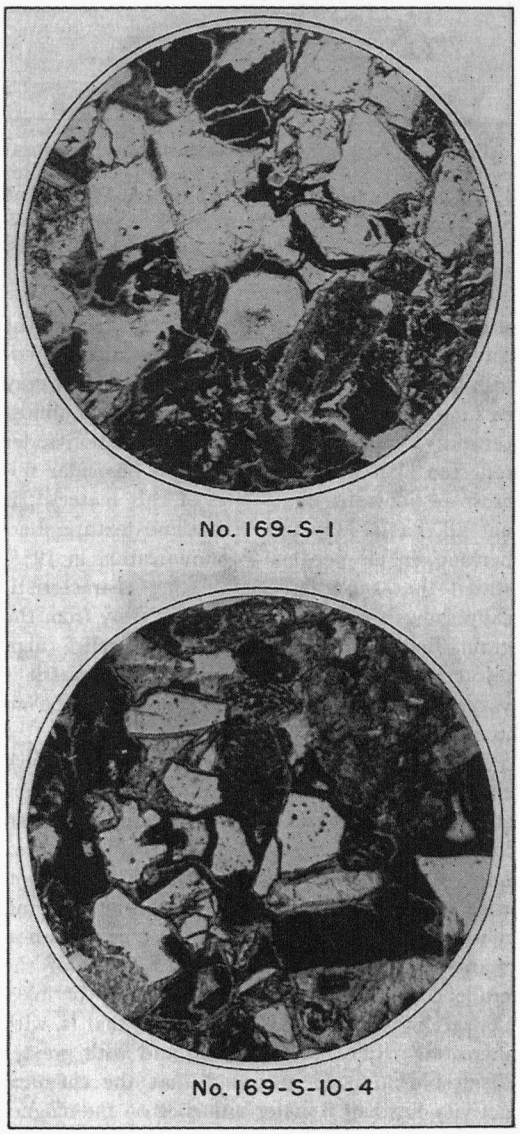


FIG. 10 – Photomicrographs of the Neroly sandstone; ordinary light, ×29 (Department of Geology Collection, University of California)

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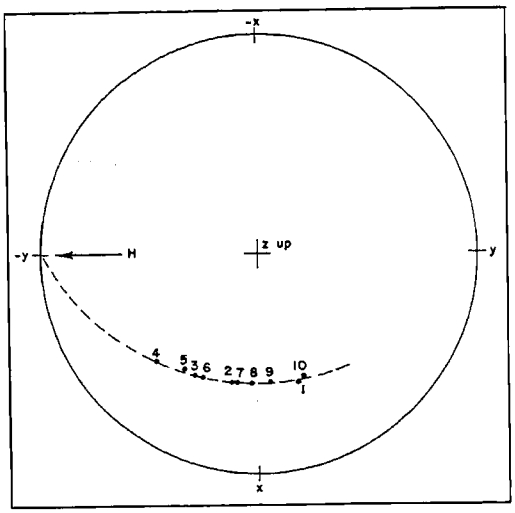


FIG. 11 – Treatment of specimen B-6-2: 1, repeat measurement; 2, 6 oe field, 15 min; 3, 6 oe field, 2 hr; 4, 6 oe field, 18 hr; 5, after 15 min recovery; 6, after 1 hr recovery; 7, after 8 hr recovery; 8, after 24 hr recovery; 9, 10 oe ac field, 30 min; 10, heated to 100°C and cooled in H = 0

stices (see Figure 10, which also clearly shows the clay coating). Some of this material appears reddish-brown along the edges, indicating that it may be limonitic or hematitic in character, and almost certainly a mineral rich in iron. Examination under reflected light and a high-powered binocular microscope indicates that some of this material, if not all, has a very fine crystalline texture. Lerbeckmo, in his personal communication in 1955, stated that the thin clay coating characteristically shows a dark rim on the side away from the grain. Because of the relatively high relief (high refractive index) of this rim, he suggests that it is very rich in iron and may quite possibly be an iron-oxide mineral.

This chemical action (deposition of the dark rims on the clay coating and filling of some interstices) might be expected to be more pronounced in the porous rocks, that is the coarser grained ones. There could also be a direct correlation with diastrophic activity, since, in general, more tightly folded rocks show greater alteration than do undisturbed sediments. Because of the crude correlation of increased intensity of magnetization with increase in grain size, that is, with increased permeability (Fig. 9) and with greater diastrophism, it is suggested that the chemical activity has had a major influence on the magnetization of these rocks, and that the detrital grains contribute but little to the total magnetization.

Effects of heat and applied fields—It is known that some rocks may possess a soft component of magnetization [*Clegg* and others, 1955; *Graham*, 1954]. This magnetization is imparted to the rock by the action of an applied unidirectional magnetic field, and is, therefore, isothermal-remanent magnetization. The possibility exists that the magnetization measured in the blue sandstones is of this type and that it was induced in these rocks by the action of the Earth's magnetic field. If this were so, then it might be expected to have the properties of IRM, in particular, a linear decrease of intensity of magnetization with temperature [*Neel*, 1955] and not a 'Curie law' decrease. A linear decrease of magnetization is not, of course, restricted to IRM, since the TRM of some spinels decreases linearly with temperature.

Specimens B-6-2, BC-5-2, and SC-1-3 were placed in dc magnetic fields of 6 oe directed along one of the cube edges in each case, and the resultant total magnetization measured after varying periods of time. The specimens were then removed from the field and the decay of this isothermal magnetization was measured. Figures 11, 12, and 13 show the effect of this treatment on the direction of total magnetization. Since the direction of magnetization in all these experiments follows a great circle containing the original direction of magnetization and the direction of the applied field, it is apparent that the components of magnetization normal to the applied field were not affected by the treatment. Further

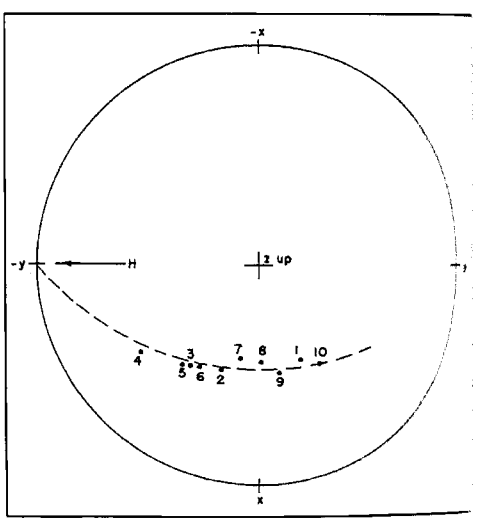


FIG. 12 – Treatment of specimen BC-5-2: 1, repeat measurement; 2, 6 oe field, 15 min; 3, 6 oe field, 1¼ hr; 4, 6 oe field, 18 hr; 5, after 15 min recovery; 6, after 1 hr recovery; 7, after 8 hr recovery; 8, after 24 hr recovery; 9, 10 oe field, 30 min (ac); 10, heated to 100°C and cooled in H = 0

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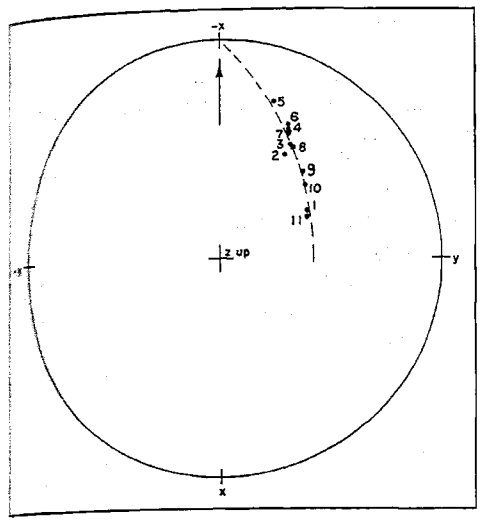


FIG. 13 – Treatment of specimen SC-1-3: 1, repeat measurement; 2, 6 oe field, 15 min; 3, 6 oe field, 45 min; 4, 6 oe field, 2 hr; 5, 6 oe field, 19 hr; 6, after 15 min recovery; 7, after 1 hr recovery; 8, after 5 hr recovery; 9, after 26 hr recovery; 10, 10 oe ac field, 30 min; 11, heated to 100°C and cooled in H = 0

after a period of relaxation of 24 hours or more, a component of isothermal magnetization remained in the direction of the applied field. This component could then be partially removed by the application of an ac magnetic field (about 15 oe) and completely removed by heating the specimen to 100°C and cooling it in a field of zero intensity. Presumably a longer application of the ac field would completely remove this isothermal magnetization also.

Several of the remaining specimens were then remeasured, heated to 100°C and cooled in zero magnetic field and measured again. A few were then heated to 200°C and cooled in zero magnetic field and again remeasured. Table 4 gives a tabulation of the intensities of magnetization determined by these measurements.

Although several of the samples increased their magnetization after heating to 100°C the general trend seems to be a decrease. To ascertain whether this decrease at 100°C is significant, the data were converted to percentage loss in magnetization, and the average decrease and probable error of this decrease were computed. The result is an average decrease in magnetization of 9½ pct. However, the probable error is 13½ pct; thus, this decrease in magnetization has no significance and is ascribed to random causes. The average loss after heating to 200°C and cooling in zero applied field is 62 pct with a probable error of 9 pct.

The test of dispersion on a sphere by *Fisher* [1953] was also applied to the directions both before and after heating to 100°C. The measure of confidence, θ, was 5½° before treatment and decreased to 4½° after heating. There is no significant difference in the average direction of magnetization between measurements before and after heating to 100°C. This is a further indication that the magnetization removed by the 100°C heat treatment was of a random nature. From these tests it is concluded that the principal

TABLE 4 – *Effects of heating*

No.	Repeat measurement ($\times 10^3$)	Heated to 100°C; cooled in H = 0 ($\times 10^3$)	Heated to 200°C; cooled in H = 0 ($\times 10^3$)
	cgsu/cc	cgsu/cc	cgsu/cc
B–1–a	9.56	11.4	...
B–2–a	24.5	22.0	...
B–3–a	39.3	39.6	22.6
B–3–c	36.6	32.7	...
B–4–a	29.7	25.0	...
B–5–1	34.4	34.9	19.6
(not included above because of lack of orientation)			
B–6–1	37.4	28.0	...
B–7–a	22.5	18.3	...
B–7–b	21.6	19.2	...
B–8–1	18.1	23.0	...
B–9–a	21.8	19.9	...
B–9–b	23.2	17.4	...
B–10–c	21.1	19.8	10.7
B–10–d	21.1	21.3	5.9
B–11–b	30.6	28.9	...
B–11–c	28.2	22.9	8.96
B–12–1	35.1	23.3	10.9
BC–1–1	13.0	8.9	4.4
BC–1–2	12.9	10.0	...
BC–2–1	5.4	4.0	...
BC–2–2	7.6	4.4	...
BC–3–1	18.2	15.7	...
BC–3–2	18.7	17.4	...
BC–4–1	18.2	14.5	...
BC–4–2	18.5	13.3	...
BC–5–1	30.6	27.6	...
BC–5–2	29.6	27.6	...
BC–6–a	6.5	5.7	...
BC–7–b	13.8	12.4	2.4
SC–1–1	14.8	13.0	...
SC–1–2	11.2	9.2	...
SC–3–1	8.2	9.3	...
SC–5–1	7.6	7.0	...
SC–5–2	8.2	6.8	...
SC–6–2	3.4	5.3	...
SC–7–1	4.2	5.0	...
SC–7–2	9.7	7.4	...
SC–8–1	2.7	2.4	...
SC–8–2	2.3	2.9	...
SC–9–2	3.0	1.6	...
SC–10–1	7.2	6.1	...
SC–10–2	7.1	6.0	...

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magnetization of these sediments is not of an isothermal type.

All the samples heated to 200°C showed marked chemical changes after this treatment. In particular, a thick brown coating formed on the surface of the originally light blue rocks and within the specimens the color had changed to a dark grey. The result was the same whether the rocks were heated in air or in an atmosphere of oil-free helium. The inference here is that the loss in magnetization is directly connected with these chemical changes. If the magnetization of these rocks had been caused by an isothermal magnetization of the detrital magnetite grains, a linear decrease with temperature should be observed instead of the marked decrease between 100° and 200°C. If the magnetization were caused by a TRM of the detrital magnetite grains, it should not be decreased appreciably at such a low temperature, since magnetites low in titanium have Curie points above 500°C [*Pouillard*, 1950]. It is possible that the detrital magnetite grains may be oxidized by the 200°C treatment. However, *Ovchinnokov* [1953], found that magnetite is stable to 220°C in an atmosphere of oxygen, which is a more oxidizing environment than that in which these sediments were heated, and that it transforms to maghemite between 220° and 260°C and goes directly to hematite above 260°C. It is therefore unreasonable to expect a chemical change of this nature to take place in the detrital magnetite grains during the heating experiments.

Conclusions—These studies seem to indicate several major points of interest:

(1) The magnetization of the rocks studied in this report is not, in the main, caused by either an alignment of the thermo-remanent magnetization or an isothermal-remanent magnetization of the detrital magnetite grains.

(2) This is a clear case of a stable magnetization in sediments that was acquired after folding and not at the time of deposition.

(3) This magnetization is not an isothermal magnetization in the usual sense, because: (a) it does not decrease linearly with temperature; (b) isothermal magnetization can be induced in these rocks and subsequently removed without affecting the original magnetization.

(4) A direct correlation may be made between the intensity of magnetization of the rocks and post-depositional chemical activity, which involves iron.

(5) A magnetization acquired as a result of chemical changes appears to have stability.

(6) There were no reversals of the Earth's field during the period(s) these rocks received their magnetization (which could have been at any time since the close of the Miocene, but was most probably during the lower Pleistocene).

On the basis of this evidence, the magnetization of these rocks is thought to be the result of chemical deposition of iron-rich minerals that took place very late in the history of the rocks. The coating itself does not have a magnetic moment, although it is somewhat paramagnetic (noted by J. Lerbeckmo, personal communication, 1955) but it is possible that the dark rims on the coating and the material filling the interstices may have. Another possibility lies in low-temperature changes on the surface of the detrital magnetite grains, but again the magnetization would be the result of a late chemical activity.

For lack of any evidence to the contrary, it seems reasonable to expect a magnetization obtained by chemical deposition (or crystal growth) to be oriented in the direction of the field acting at the time of deposition. 'Crystallization magnetization' is the name suggested for this phenomenon, which may be summarized as follows. A crystal growing or crystallizing in a weak magnetic field may acquire a magnetization that is more intense and more stable than the magnetization this same crystal (originally unmagnetized) would acquire if it were placed when fully grown in a field of the same intensity at the same temperature. There is no known experimental evidence to confirm this. If the crystals do not grow to a size large enough to consist of more than one domain (which is probably the case for the rocks studied here), the direction of magnetization of this domain might be expected to be in the direction of the applied field. A single-domain grain might also automatically develop with its direction of easy magnetization parallel to the applied field, since this orientation would require much less energy than would be needed later to turn the magnetization of such a single-domain grain against the anisotropy energy. Further, if the grains become larger than those consisting of one domain, it seems reasonable to expect an easier displacement of the domain boundaries while the crystal is growing than after it is fully grown.

To what extent this process takes place in other rocks is not known. However, the possibility should not be excluded, especially if the rocks in question come from a folded region or show evidence of a post-depositional chemical activity.

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